1.2 Measuring and Constructing Segments

Segment length is a number, not just a picture — and the Segment Addition Postulate and the Midpoint definition turn that number into an algebra tool. This section goes past single-step "add the two pieces" problems into equations with a variable on both sides, midpoints of midpoints, and word problems where you must first decide which points are the endpoints before any arithmetic happens. Expect to justify each step (postulate or definition) as much as to compute the final number.

Examples

Example 1: Distance on a number line

Find AB if A is at -3 and B is at 5 on a number line.

Solution



Distance is the positive gap between coordinates.

Identify the formula: distance between two points on a number line is $|x_2 - x_1|$ (order doesn't matter because of the absolute value).
Substitute: $AB = |5 - (-3)| = |8|$.
Answer: $AB = 8$.

Example 2: Segment Addition Postulate

B is between A and C. If $AB = 4$ and $BC = 7$, find AC.

Solution



B is between A and C, so the two small pieces add to the whole.

Identify the relationship: since B is between A and C, the Segment Addition Postulate applies: $AB + BC = AC$.

Substitute the known values: $AC = 4 + 7$.

Answer: $AC = 11$.

Example 3: Find a missing part

B is between A and C. If $AC = 20$ and $AB = 8$, find BC.

Solution

Start from the same postulate, but solve for the middle piece instead of the whole: $AB + BC = AC$.

$8 + BC = 20$

$BC = 20 - 8$.

Answer: $BC = 12$.

Example 4: Midpoint doubles the half

M is the midpoint of $\overline{AB}$. If $AM = 9$, find AB.

Solution



M splits AB into two equal pieces, AM and MB.

Identify: a midpoint splits a segment into two *congruent* pieces, so $AM = MB$.
Since $AM = 9$, also $MB = 9$; then $AB = AM + MB = 9 + 9$.
(Shortcut for any midpoint problem: $AB = 2 \cdot AM$.)
Answer: $AB = 18$.

Example 5: Midpoint algebra

M is the midpoint of $\overline{PQ}$. If $PM = 3x + 1$ and $MQ = 5x - 5$, find x.

Solution

Set the two halves equal, since a midpoint makes them congruent: $PM = MQ$.
$3x + 1 = 5x - 5$
$1 + 5 = 5x - 3x \Rightarrow 6 = 2x$
$x = 3$.
Check: $PM = 3(3) + 1 = 10$ and $MQ = 5(3) - 5 = 10$ — equal, as required. ✓

Example 6: Congruent segments and transitivity

$\overline{AB} \cong \overline{CD}$ and $\overline{CD} \cong \overline{EF}$. If $AB = 12$, find CD and EF, and name the property that lets you compare AB and EF directly.

Solution

Definition of congruent segments: congruent means equal length. $\overline{AB} \cong \overline{CD} \Rightarrow CD = AB = 12$.

$\overline{CD} \cong \overline{EF} \Rightarrow EF = CD = 12$.

Since $AB = CD$ and $CD = EF$, the **Transitive Property of Equality** gives $AB = EF$ without re-measuring.

Answer: $CD = 12, EF = 12$.

Example 7: Solve with Segment Addition

B is between A and C. $AB = 2x$, $BC = x + 4$, and $AC = 19$. Find x.

Solution



The two algebraic pieces must add to the given whole.

Write the postulate as an equation: $AB + BC = AC$.

$2x + (x + 4) = 19$

$3x + 4 = 19 \Rightarrow 3x = 15 \Rightarrow x = 5$.

Check: $AB = 10, BC = 9$, and $10 + 9 = 19$. ✓

Example 8: Midpoint coordinate on a number line

A is at 2 and B is at 14. Find the coordinate of the midpoint M.

Solution



The midpoint coordinate is the average of the two endpoints.

Formula: the midpoint coordinate is the average of the endpoints, $M = \dfrac{x_A + x_B}{2}$.

$M = \dfrac{2 + 14}{2} = \dfrac{16}{2}$.

Answer: $M = 8$.

Example 9: Find a missing endpoint

M is the midpoint of $\overline{AB}$. A is at 4 and M is at 10. Find B.

Solution

Use the midpoint formula, but solve for the unknown endpoint instead of the midpoint: $M = \dfrac{A + B}{2}$.

$10 = \dfrac{4 + B}{2} \Rightarrow 20 = 4 + B$

$B = 16$.

Check: the midpoint of 4 and 16 is $\dfrac{20}{2} = 10$. ✓

NAME: _____ Geometry (FL B.E.S.T.)

Example 10: Nested midpoints — a hallway word problem

A water fountain F is at mile-marker 5 and a door D is at marker 35 along a straight hallway. A bench B is placed at the midpoint of $\overline{FD}$, and a trophy case T is placed at the midpoint of $\overline{FB}$. Find the coordinates of B and T, and find FT.

Solution

Find B first, since T depends on it: $B = \dfrac{F + D}{2} = \dfrac{5 + 35}{2} = \dfrac{40}{2} = 20.$

Now find T, the midpoint of $\overline{FB}$: $T = \dfrac{F + B}{2} = \dfrac{5 + 20}{2} = \dfrac{25}{2} = 12.5.$

Finally, $FT = |T - F| = |12.5 - 5| = 7.5.$

Answer: $B = 20$, $T = 12.5$, $FT = 7.5$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Distance

A at -2, B at 6. Find AB.

Answer: **Verification:**

Problem 2: Segment Addition

B between A and C, $AB = 5$, $BC = 9$. Find AC.

Answer: **Verification:**

Problem 3: Find a part

$AC = 25$, $AB = 11$, B between A and C. Find BC.

Answer: **Verification:**

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Problem 4: Midpoint length

M midpoint of $\overline{AB}$, $AM = 7$. Find AB.

Answer: **Verification:**

Problem 5: Midpoint algebra

$PM = 2x + 3$, $MQ = 4x - 5$, M midpoint. Find x.

Answer: **Verification:**

Problem 6: Congruent segments

$\overline{RS} \cong \overline{TU}$ and $RS = 15$. Find TU.

Answer: **Verification:**

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Problem 7: Solve for x

$AB = x$, $BC = 2x + 1$, $AC = 16$, B between. Find x.

Answer: **Verification:**

Problem 8: Midpoint coordinate

A at 3, B at 11. Midpoint M?

Answer: **Verification:**

Problem 9: Endpoint

A at 1, midpoint M at 5. Find B.

Answer: **Verification:**

Problem 10: Distance

P at -8, Q at -1. Find PQ.

Answer: **Verification:**

Problem 11: Segment Addition

$AB = 3x - 1$, $BC = x + 5$, $AC = 20$. Find x.

Answer: **Verification:**

Problem 12: Midpoint length

$MQ = 6$, M midpoint of $\overline{PQ}$. Find PQ.

Answer: **Verification:**

Problem 13: Midpoint algebra

$AM = 5x - 2$, $MB = 3x + 6$. Find x.

Answer: **Verification:**

Problem 14: Challenge — nested midpoints

Trail markers: A at mile 3, D at mile 27. B is the midpoint of $\overline{AD}$, and C is the midpoint of $\overline{BD}$. Find B, C, and BC.

Answer: **Verification:**

Problem 15: Challenge — Segment Addition plus midpoint

B is between A and C. $AB = 2x + 3$, $BC = 3x - 1$, and $AC = 32$. Find x, then find AM where M is the midpoint of $\overline{AC}$.

Answer: **Verification:**

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✓ Answer Key

1. 8
2. 14
3. 14
4. 14
5. $x = 4$
6. 15
7. $x = 5$
8. 7
9. 9
10. 7
11. $x = 4$
12. 12
13. $x = 4$
14. $B = 15, C = 21, BC = 6$
15. $x = 6; AC = 32; AM = 16$

Great work — keep practising!